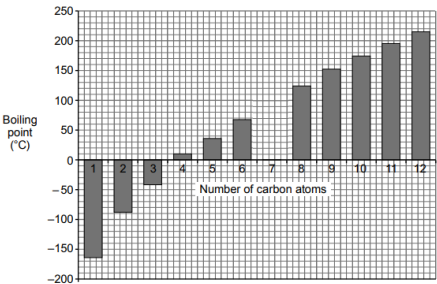
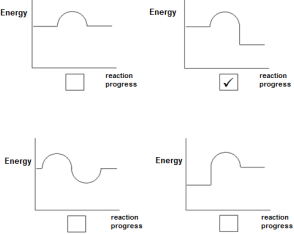


Common questions

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6/1	(a)			the <u>larger the molecules / (as the size/fraction) gets bigger</u> - the <u>smokier the flame</u> (1) - the <u>more difficult</u> it becomes to <u>burn</u> (1) accept the converse argument correct identification of both properties, without reference to increasing size (1)			2	2		
	(b)	(i)		 all 5 bars correctly plotted with $\pm\frac{1}{2}$ small square tolerance (2) 3 or 4 correct plots (1) accept charts where bars are touching – correct height of each bar to be credited		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		appropriate straight trend line drawn with ruler, spans across at least 6 bars (1) correct boiling point taken from the trend line (1) award no credit for a curved trend line (but allow ECF for a correct value read from an incorrect trend line) if no trend line drawn, award (1) for a value in the range 85-105			2	2	2	
	(c)	(i)		sand / foam / CO ₂ / fire blanket because it removes oxygen both method and explanation needed		1		1		
		(ii)		5CO ₂ + 6H ₂ O correct products (1) correctly balanced (1) – only if correct products given		2		2	1	
		(iii)		does not produce carbon dioxide / sulfur dioxide / only produces water (1) does not contribute to global warming / climate change / acid rain (1)	1	1		2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		temperature (of water) before and after burning (1) mass of fuel and burner before and after burning (1) initial $\equiv$ before $\equiv$ at start final $\equiv$ after $\equiv$ at end temperature rise and change in mass – neutral answers award (1) only for reference to measuring both temperature throughout and mass throughout award (1) for answers that refer to measuring both the temperature and mass either before or after burning only			2	2		2
		(ii)		distance between burner and flame / material or thickness or size of beaker / same size wick / same beaker reference to mass of fuel and mass of water – neutral	1			1		1
		(iii)				1		1	1	
				Question 6/1 total	2	7	6	15	6	3